# STAT 330-001

40142034 52

## Strang, Ethan Tyler

027821042 strange 661-373-8227

Test: 641136 Form: A Wadsworth, Spencer Gordon

Entry: 05:56pm 10/21/2026 Checked out:

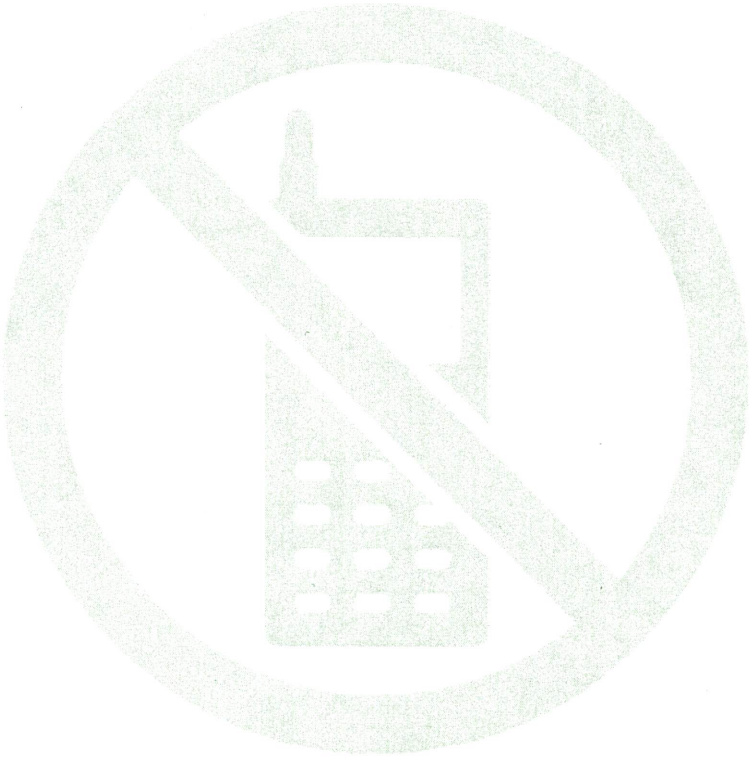
Employee: Myles R.

**SPECIAL INSTRUCTIONS**

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

## PLEASE TURN OFF YOUR CELL PHONE



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052

# STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

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- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
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## PART I

**Question 1: (Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

**Question 2: (TRUE/FALSE)** If the relationship between $x$ and $y$ is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
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**Question 3: (Circle one answer)** When adding an additional variable to a linear regression model, the model's $R^2$ ...

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**Question 4: (Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination ($R^2$) to become negative.

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(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the $x$'s or $y$ may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of $\beta$ can be obtained by fitting a simple linear regression for each $x_k$ only.

(E) The assumptions on the error variables (i.e., the $\epsilon_i$) in a multiple linear regression model are exactly the same as they are for simple linear regression.

**Question 6: (Circle one answer)** What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

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**Question 7: (Circle all that apply)** In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

These methods shrink the $\hat{\beta}_i$'s. LASSO uses $L_1$ $\left(\sum_{i=1}^{\hat{}} |\beta|\right)$, while Ridge uses $L_2$ $\left(\sum_{i=1}^{\hat{}} \beta^2\right)$. Some values for LASSO even go to zero.

$\frac{5.76}{4} \cdot \left(\frac{1}{3}\right) = .48$

1.44 $p+1 = 4$

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right)$$

$\left(\frac{2.4^2}{3+1}\right)\left(\frac{.25}{1-.25}\right)$

**(Circle one answer)** Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

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(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

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(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

**Question 10:**

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = (X^T X)^{-1} X^T Y$

(B) What is the conceptual difference between $\widehat{\beta}$ and $\beta$?

$\hat{\beta}$ is the numerical estimate obtained from the data. It helps estimate how much a predictor changes the response variable.

$\beta$ is used to create theoretical models (i.e. $Y = \beta_0 + \beta_1 x + \varepsilon$) and is not a # (yet).

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

$\sum_{i=1}^{n} e_i^2$ ~~$SSE(\beta) = (Y - X\beta)^T (Y - X\beta)$~~ ~~$\rightarrow X^T X \hat{\beta} = X^T Y$~~

least squares

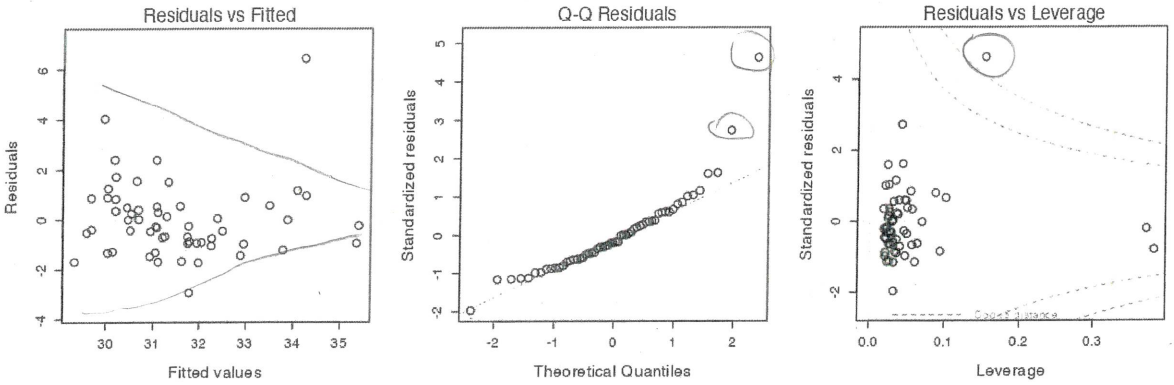
$e_i = Y_i - \hat{Y}_i$ It subtracts the mean/best $\hat{Y}_i$ by the actual $Y_i$

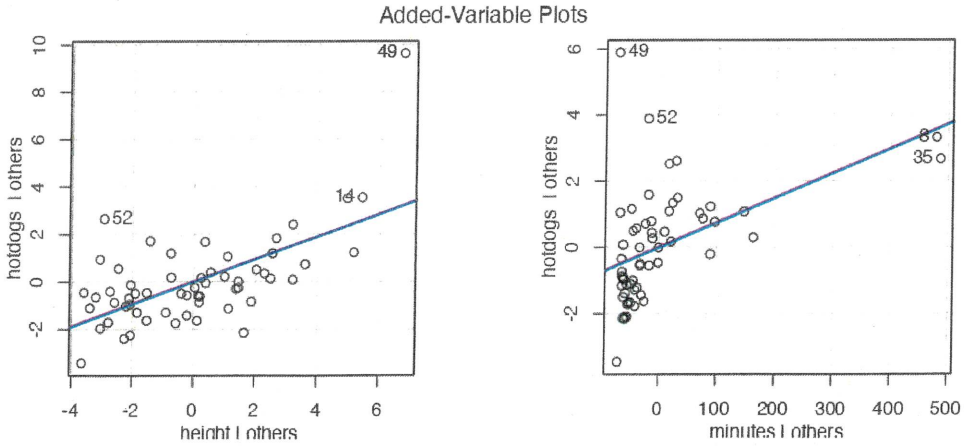
# PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

$\beta_1 = 0.472892$ Holding minutes constant, a 1 inch increase for a contestant is predicted to increase the amount of hot dogs eaten in the competition by 0.472892.

**Question 12:** **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0 estimate $\pm$ SDE $\cdot t^*$

(B) The 95% confidence interval of $\beta_2$ does contain 0

(C) There is an ~~81.4%~~ probability that $\beta_0 = 0$ $.007417 \pm .001871 \cdot t^*_{3,963}$

(D) $\beta_1 = 0.4729$ $= [0.0148317 \leftrightarrow 0.0000227$

(to find the variance $\beta_2$)

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

In the Residuals vs Fitted Plot, I see a funnel \ indicating that it is not linear.

(B) There is independence between each observation

The # of hot dogs and the height variable are independent, but I am worried about the $\beta_2$ Minutes variable. The contestants were asked the minutes question, which could result in untruthful responses. If they were all asked at once in a crowd, answers would be effected by social pressures.

(C) The errors are normally distributed I am simply unsure if $\beta_2$ is independent

The Q-Q plot has a bow that needs to be addressed. If the 2 furthest away points were gone, it would be normal. Those points circled should be investigated

(D) The errors have constant variance (homoscedasticity)

Since P value is high for the Breusch-Pagan test, errors could have constant variance. We don't have a scale location plot to visual determine homoscedasticity.

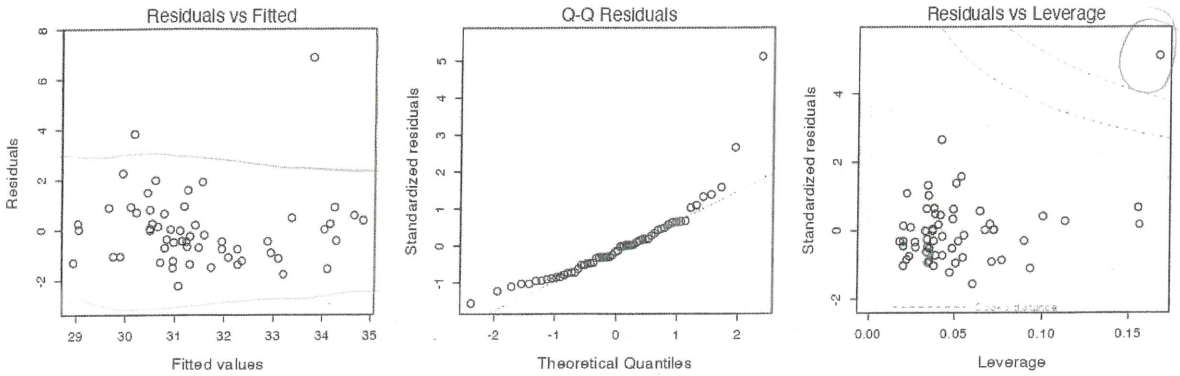
(E) There are no unduly influential points

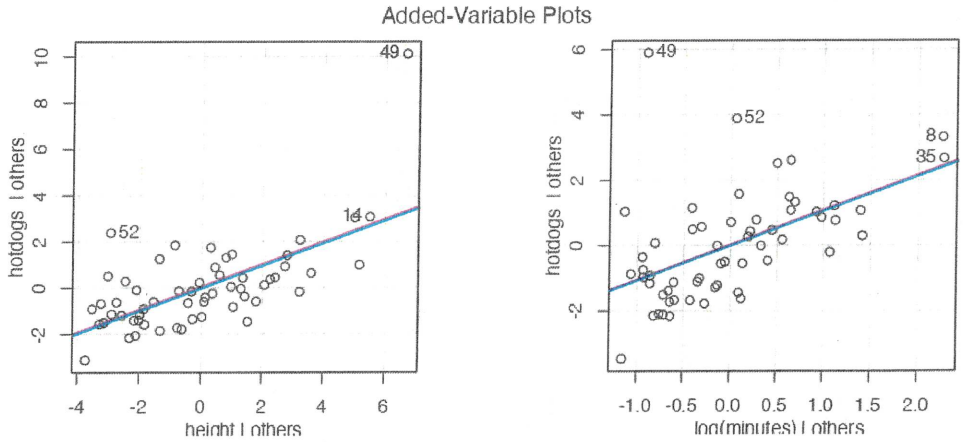
The Residual vs Leverage plot shows 1 point beyond the rule of thumb of 5 (Post — Jing) This needs to be ~~done~~ checked out.

## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| log(`minutes`) | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

Looking at the Breusch-Pagan test, I would agree. I would tell the researcher that it is not super clear that model 1 is superior due to the LINE worries we had. Linearity and the rouge values are big issues to deal with before declaring a better model.

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Overall, they appear similar. An advantage is that it appears more linear from the risidual vs fitted plot. However, the lower p-value in the Breusch-Pagan test is a disadvantage.

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context

c% $\log\left(1+\frac{c}{100}\right)$

Holding height constant, a 1 minute increase of training is predicted with an increase of about 1.016702 hotdogs eaten.

$\left[\log\left(1+\frac{c}{100}\right)\%, \log\left(1+\frac{1.016702}{100}\right)\%, \log(1.016702) \approx 1.016702\right]$

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that $\log(79) = 4.369$. Use this information and the fit for model (2) to answer the following questions.

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\widehat{\text{hotdogs}} = -6.05559 + 0.41334(\text{inches}) + 1.06702(4.369)$

$= 31.813$ hot dogs

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

CI

interval = estimate $\pm$ $t^*$ · SE

$1.485 \pm t^* \cdot 0.204$

$t = 1.29$??

$- = 1.076124$

$.204$

$1.06702 \pm 2.004$ (.2407)

$[0.6582 \quad 1.475836]$ for .204

$[-1.9089 \quad 4.04296]$ for 1.485

# Part III - Application

## Questions 20-24 concern the following dataset:

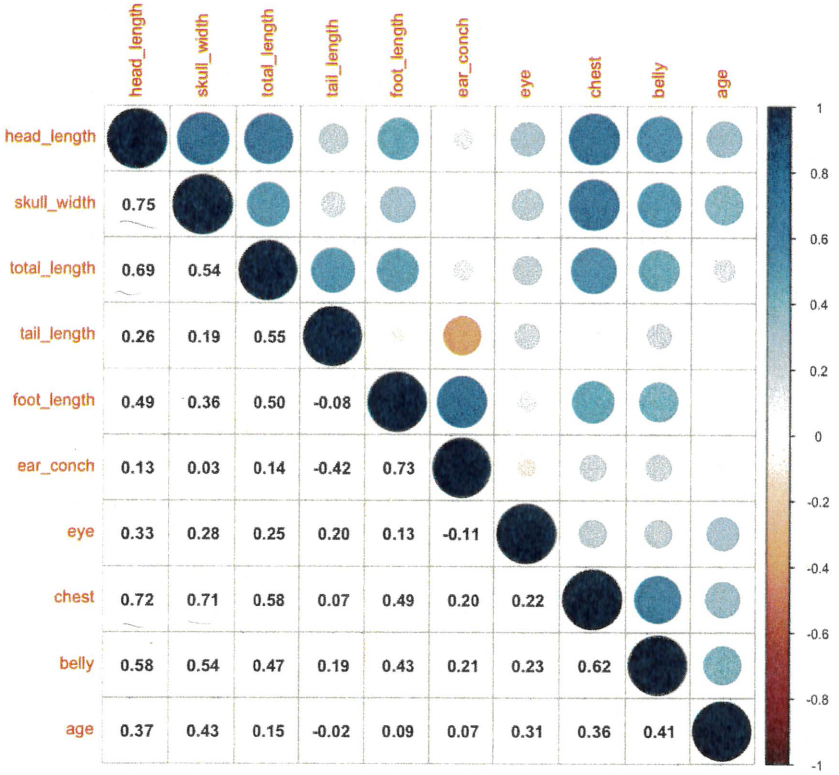
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

| Variable | Description |
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
foot_length   -0.21499    0.10733  -2.003   0.0507 .
ear_conch      0.15730    0.10231   1.538   0.1306
eye            0.43736    0.22549   1.940   0.0582 .
chest          0.01883    0.18154   0.104   0.9178
belly          0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

estimate $\pm$ t* SE

$0.16559 \pm (2.01 \cdot .10599)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the true mean of the belly width coefficient is between $-0.023$ and $0.443$

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

Multicolinearity is the reason for the "contradiction." The predictor variables may have an effect on each other. This creates a situation where the model works, but it is difficult to identify what each variable is doing in specific. It is overfitted.

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

$n \log(SSE/n) + 2k$

The "best subset" algorithm works through $2^n$ model combinations to find the best fit. The AIC punishes added variables to the model by $2k$ in order to get the best model.

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

B Average the predictions and record their value A B

D Choose the model with the largest RMSE A) in this case it did not happ

A Choose the model with the smallest RMSE A

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

B Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

B Initialize vectors for each model that you will use to store their RMSEs

C Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

# STAT 330-001

40142034 51

## Gray, Zane Tellis

288433465 zanetgry 626-433-7107

Test: 641107 Form: A
Entry: 05:49pm 10/21/2026
Employee: Ainsley C.

Wadsworth, Spencer Gordon
Checked out:

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(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

They shrink the predictor coefficients by applying penalties such as $\lambda \sum |B_i|$ for lasso and $\lambda \sum B_i^2$ for ridge.

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

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$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

**(Circle one answer)** Which conclusion is most appropriate?

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(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

**Question 10:**

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = (X^TX)^{-1}X^TY$

(B) What is the conceptual difference between $\widehat{\beta}$ and $\beta$?

$\hat{\beta}$ is calculated on data that has been collected and then fit On the other hand $\beta$ is the theoretical model to predict Y.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

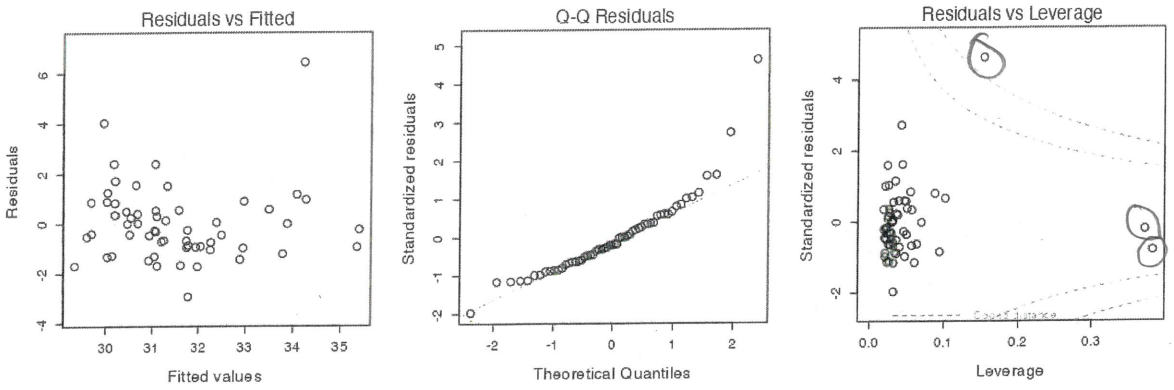
When finding a regression coefficient you want to minimize the sum of square, Thus, using least squares provides us with the solution to finding the coefficients with the smallest sum of squares.

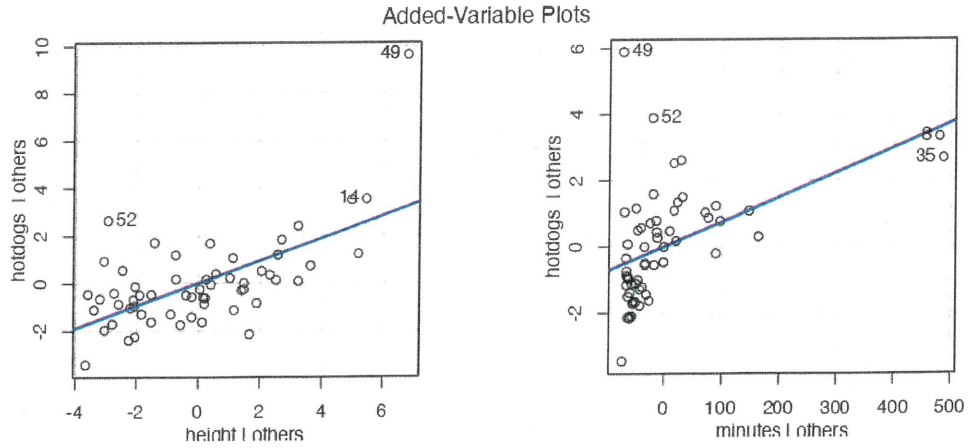
# PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | Pr($>\|t\|$) |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

$\beta_1 = 0.472892$

When holding minutes of training constant, every 1 unit increase in height will produce an increase of 0.472892 hot dogs consumed during the ten minute competition.

**Question 12:** **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0

(B) The 95% confidence interval of $\beta_2$ does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

$\beta_2 - t^*(SE) \approx 0.00000223$

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Given the Residuals vs Fitted and Added-Variable plots, I think it is safe to assume they are linear despite some concern with the hotdog and minutes Added-Variable plot.

(B) There is independence between each observation

There could be some concern with independence given that a majority of the participants were from just 4 counties but besides that it is safe to assume independence.

(C) The errors are normally distributed

Given the QQ-plot and its bowed shape along with the significantly low p-value for the Shapiro-Wilk test, I don't think this model's errors are normally distributed.

(D) The errors have constant variance (homoscedasticity)

We see that the Breusch-Pagan test resulted in a p-value greater than 0.05 and the Residuals vs Fitted plot looks good. Thus, it is safe to assume equal variance and no heteroscedasticity.

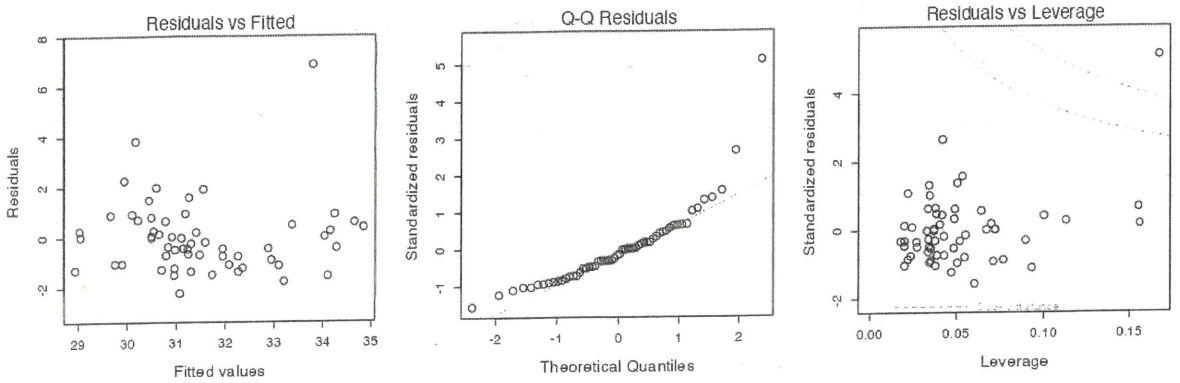
(E) There are no unduly influential points

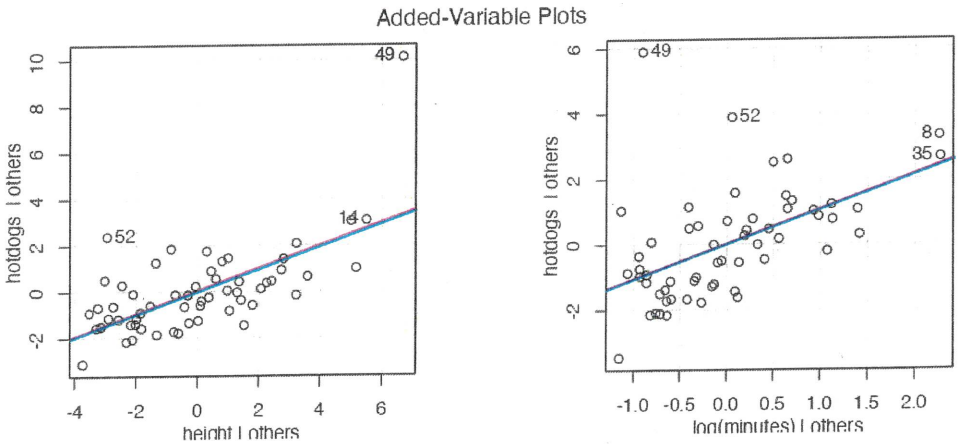
Given the Residuals vs Leverage plot, I would be mildly concerned with the three outliers I have circled, but no need to discard them until further investigation.

## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| log(`minutes`) | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would agree that the result of the Breusch-Pagan test is concerning. However, in Model 2 there are clear improvements to the QQ and Added-Variable plots. Thus, there is reason to use Model 2.

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

As explained above, among improvements in a majority of the plots the CV RMSE also decreased in Model 2 compared to Model 1.

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context

$\log(\beta_2) = 1.06702$

Holding height constant, a one-percent increase in minutes trained will produce a 0.0106702 increase in hot dogs consumed during the ten minute competition.

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that $\log(79) = 4.369$. Use this information and the fit for model (2) to answer the following questions.

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$e^{4.369} \approx 78.965$ hot dogs in 10 minutes

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$78.965 \pm 2.004(.204 + 1.485)$

PI: (75.58, 82.35)

# Part III - Application

## Questions 20-24 concern the following dataset:

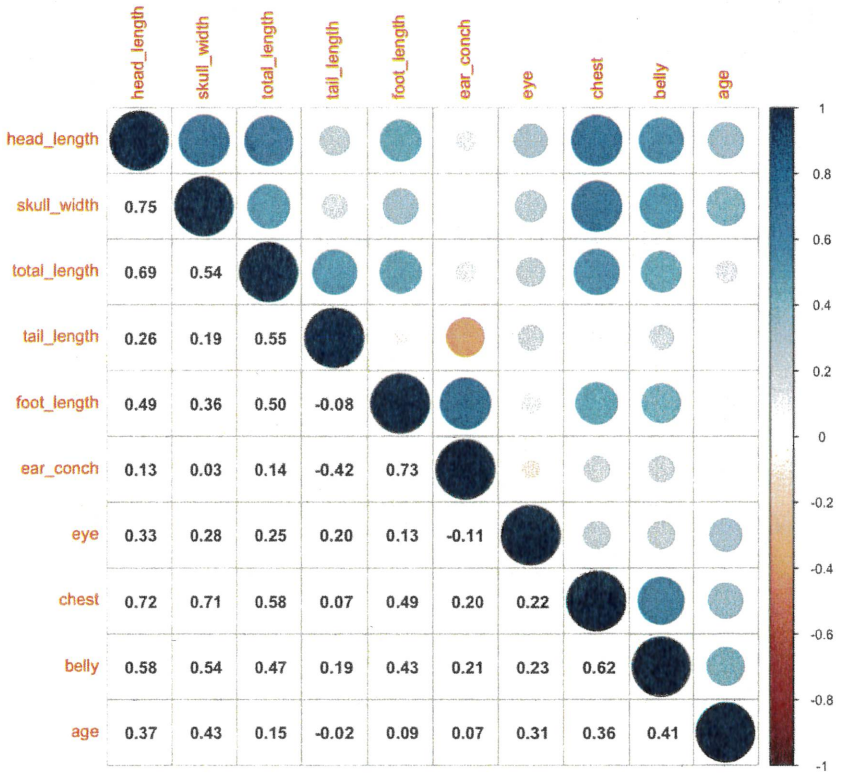
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

| Variable | Description |
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$0.16559 \pm 2.01(0.10599)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the true mean value of $\beta_{belly}$ falls between -0.023 and 0.443. Since, the interval contains zero we fail to reject $H_0$.

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

The "contradiction" is the F-test simply implies that due to its p-value being less than 0.05 we assume that at least one slope of the model coefficients is not zero.

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

The best subsets algorithm will compare combinations of predictors to find which model is best given the data. Meanwhile, AIC favors models with more variables that have smaller effects and a set penalty at $2k$ added to $2\ln(L)$, where $L$ is the maximum likelihood. However, best subsets is penalized by $2^p$ as the number of predictors increases.

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

A Average the predictions and record their value

D Choose the model with the largest RMSE

A Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

C Compute each model's AIC and BIC

B Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

A Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

C Use the models fitted to the training data to get predictions for the test data

END OF EXAM

# STAT 330-001

8*

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## Chandler, Eldon

502116119 emchand2 435-531-3379

Test: 640974 Form: A
Entry: 05:20pm 10/21/2026
Employee: Ainsley C.

Wadsworth, Spencer Gordon
Checked out:

**SPECIAL INSTRUCTIONS**

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

**PLEASE TURN OFF YOUR CELL PHONE**

5 ✓
8 —
10 ✓
12

40142034

050

# STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

**Instructions**

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

## PART I

**Question 1: (Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

**Question 2: (TRUE/FALSE)** If the relationship between $x$ and $y$ is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

**Question 3: (Circle one answer)** When adding an additional variable to a linear regression model, the model's $R^2$ ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

**Question 4: (Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination ($R^2$) to become negative.

**Question 5: (Circle all that apply)** Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the $x$'s or $y$ may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of $\beta$ can be obtained by fitting a simple linear regression for each $x_k$ only.

(E) The assumptions on the error variables (i.e., the $\epsilon_i$) in a multiple linear regression model are exactly the same as they are for simple linear regression.

**Question 6: (Circle one answer)** What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

**Question 7: (Circle all that apply)** In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

They "shrink" the coefficient estimates introducing bias to have less variance in the model. Helps to prevent the model to be less affected by the "quirks" in the data

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$\frac{(2.4)^2}{4}\left(\frac{0.25}{0.75}\right)$

**(Circle one answer)** Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

1 44

=0.48

=0.044

**Question 10:**

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = (X^T X)^{-1} X^T Y$

(B) What is the conceptual difference between $\widehat{\beta}$ and $\beta$?

(least Squares estimate)

$\hat{\beta}$ is the estimate of the coefficient, & $\beta$ is the true value of that coefficient within the population

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

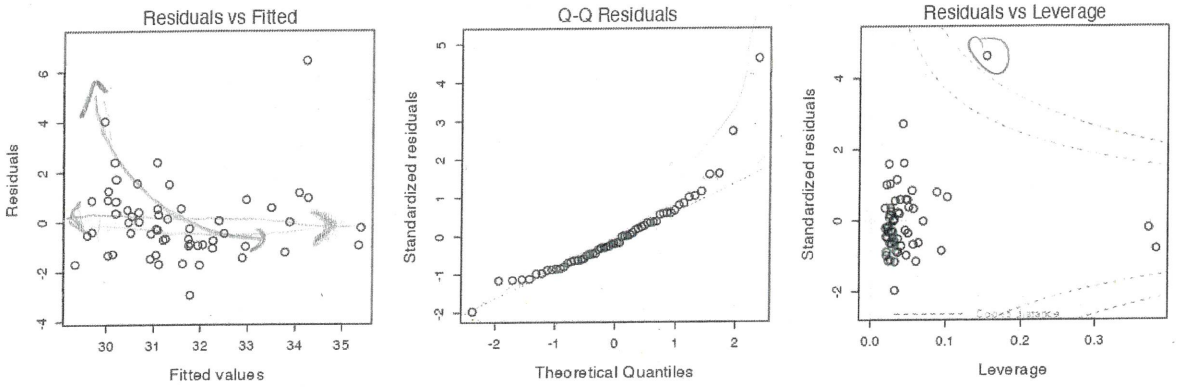
Least Squares is used By Squaring all of the residuals so they Don't Cancel each other out, if negative & positive, & Then using that TO fit A model where the Distance Between our fit & given values is minimized

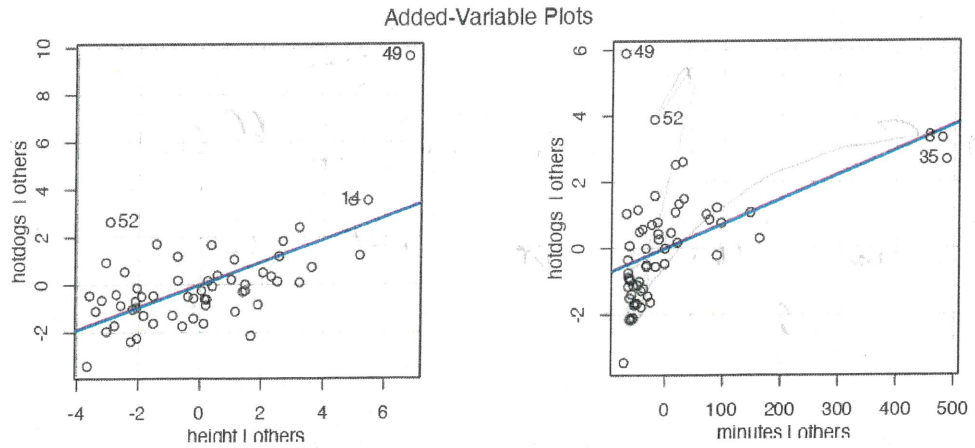
# PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | Pr($>\|t\|$) |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

0.47, Holding minutes Constant. There is an estimated increase of 0.47 Hotdogs Consumed for every 1 inch increase in Height ↓ in 10 min

**Question 12:** **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0

(B) The 95% confidence interval of $\beta_2$ does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$ ?

(D) $\beta_1 = 0.4729$ X $\hat{\beta}$

$0.0074 \pm 3.93(0.001871) = 0.007353$

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

it could be plausible, but there appear to be some likely issues. Looking at the residual vs fitted there appear to be curvature, or a downward trend, AV plot for minutes also does not appear to be a random cluster around the best fit line

(B) There is independence between each observation

Seems Reasonable, They come from the 4 largest counties, so if they are training together in the same way & sharing mentors, this could be an issue

(C) The errors are normally distributed

QQ plot shows Deviance from the normal error Distribution line, & Shapiro wilk shows a low p-val causing us to fail to Reject null ? state not enough evidence errors are normally Distributed

(D) The errors have constant variance (homoscedasticity)

There is a fan shape of funnel in the Resid vs fitted plot, the BP test gives us a p value of 0.0717 > 0.05, so CV may be plausible

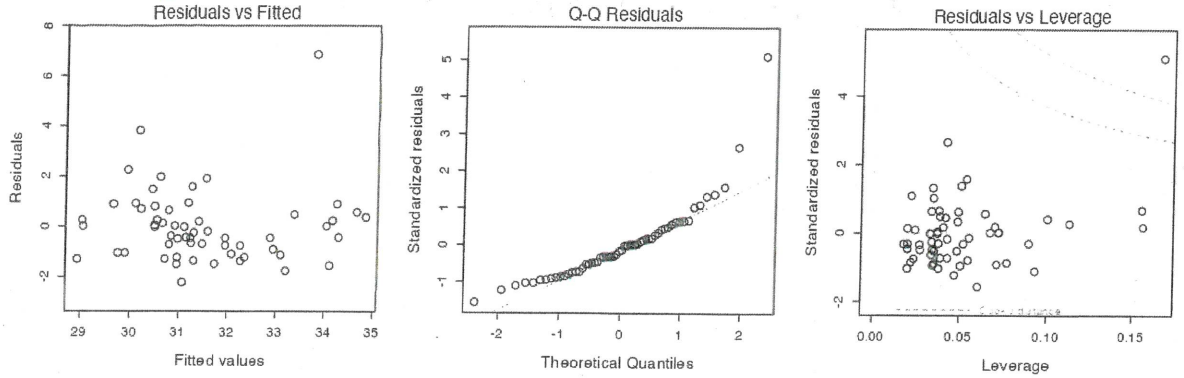
(E) There are no unduly influential points

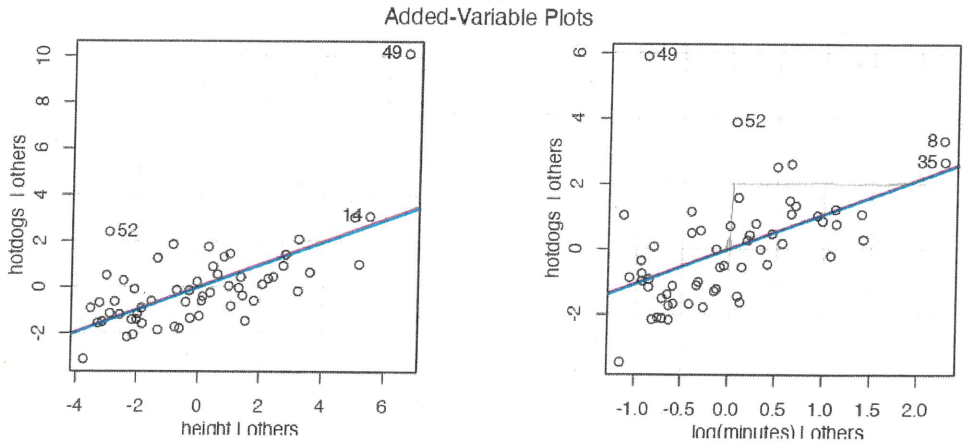
There is a point that has more influence than perferred, and it would be best to look into it to ensure it should belong in our sample

## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| log(`minutes`) | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

This is not instantly true, while A pval of 0.071 is better for CV, other factors need to be considered, and A 0.071 is not A substantial Difference from 0.028 for us to make that strong of A claim

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

MODEL 2 appears more Linear & errors are Distributed more normal From A Higher pval from Shapiro Wilk
from added variable plots random cluster

MODEL 1 has A more Constant variance
We knew this Because of A Higher pval for Bp test when Compared TO model 2

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context

per week used TO Train for Contest

Level log for A 10% increase in minutes, there is an estimated 4.4% increase in Hotdogs eaten in 10 min Holding Height Constant

1.06 $\log(1.1) = 0.04$

0

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\hat{Hotdogs} = -6.05589 + 68(0.48834) + 4.369(1.06702)$

$\approx 31$ Hotdogs

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

$58 - 2 - 1$

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$31 \pm 2.004\,(1.485)$

95% conf: $(\approx 28, \approx 34)$

# Part III - Application

## Questions 20-24 concern the following dataset:

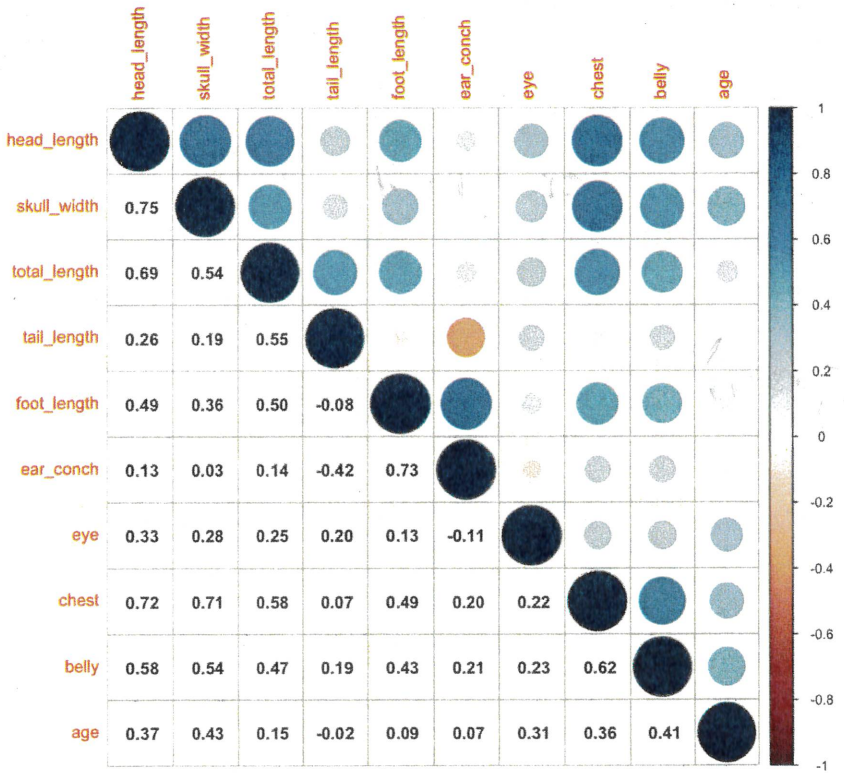
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

| Variable | Description |
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$0.16559 \pm 2.01(0.10599)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

If we were to reproduce this, the true change in years for every 1cm increase in Belly Girth would be captured within the interval (-0.023, 0.443) yrs 95% of the time

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

F test determines at least 1 predictor is linearly related with response. This can be happening because of multicollinearity within the DATA, making the F-test result unreliable

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

it is an exhaustive search & will make all possible... models from the predictors and use AIC as a metric to determine which model is optimal

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

# STAT 330-001

40142034 49

## Crawford, Spencer Matthew

091942619 crawfo22 801-319-3523

Test: 640775 Form: A Wadsworth, Spencer Gordon

Entry: 04:33pm 10/21/2026 Checked out:

Employee: Jemma C.

**SPECIAL INSTRUCTIONS**

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly.
Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

**PLEASE TURN OFF YOUR CELL PHONE**

NS

40142034

049

# STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

**Instructions**

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

## PART I

**Question 1: (Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

**Question 2: (TRUE/FALSE)** If the relationship between $x$ and $y$ is nonlinear, a simple linear regression model may still be feasible after transforming the data.

• TRUE

• FALSE

**Question 3: (Circle one answer)** When adding an additional variable to a linear regression model, the model's $R^2$ ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

**Question 4: (Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination ($R^2$) to become negative.

**Question 5: (Circle all that apply)** Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the $x$'s or $y$ may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of $\beta$ can be obtained by fitting a simple linear regression for each $x_k$ only.

(E) The assumptions on the error variables (i.e., the $\epsilon_i$) in a multiple linear regression model are exactly the same as they are for simple linear regression.

**Question 6: (Circle one answer)** What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

**Question 7: (Circle all that apply)** In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

$\beta_j$ is what is being shrunk, to better balance the predictors, this is done by assigning a penalty based on that $\beta_j$ (either $\Sigma \beta_i^2$ or $\Sigma|\beta|$) and a $\lambda$ that we select.

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$\frac{2.4^2}{4}\left(\frac{0.25}{0.75}\right) = 0.48$

$\frac{4}{100} = 0.04$

**(Circle one answer)** Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

**Question 10:**

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = X(X^TX)^{-1}X^T\hat{Y}$

(B) What is the conceptual difference between $\widehat{\beta}$ and $\beta$?

$\hat{\beta}$ is our estimated $\beta$

$\beta$ is the coeficient for our predictor variables

★(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

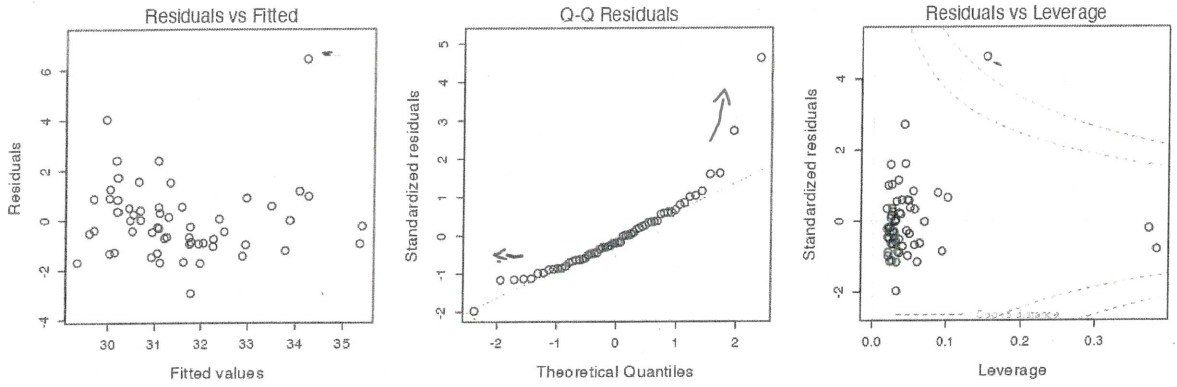
Our $\hat{\beta}$ is estimating the values with the smallest SSE, it takes the value that minimizes the sum of the error$^2$ for each coefficient

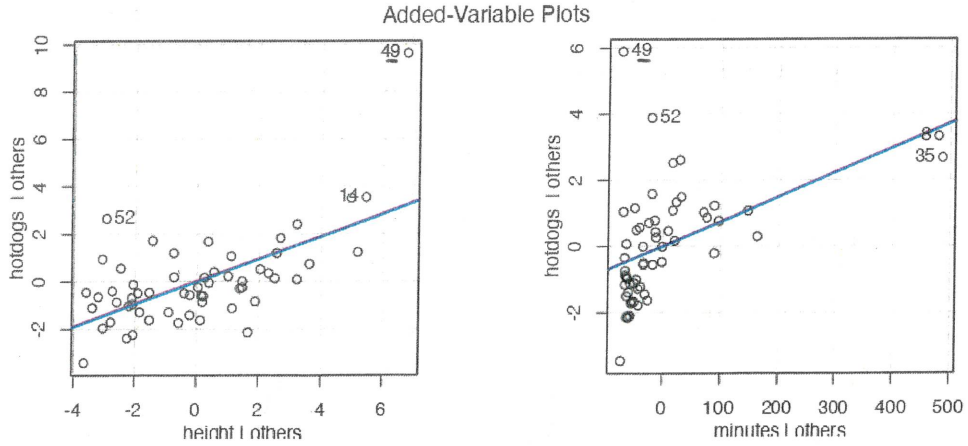
# PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | Pr($>\|t\|$) |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

0.472892

holding minutes constant we predict that on average, for every inch in height increased the amount of hot dogs eaten in 10 minutes will go up by approximately 0.472892

**Question 12:** **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0

(B) The 95% confidence interval of $\beta_2$ does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$ $\hat{B}_1 = 0.4729$

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

for the most part it is linear there are some violations where our AV plot between hotdogs/other and minutes/other does not follow the linear regression very well,

(B) There is independence between each observation

For the most part it is satisfied, as one competitor should not affect another, but there could be some dependence, such as an increase in competition if they see each other,

(C) The errors are normally distributed

Our Q-Q Plot shows quite a bit of skewness, and the Shapiro-Wilk test quite significant, we would want to investigate this one further

(D) The errors have constant variance (homoscedasticity)

Our BP test is not extremely significant, and the Residual vs fitted plot doesn't have too many issues but there is a little bit of change as it has a bit of shape

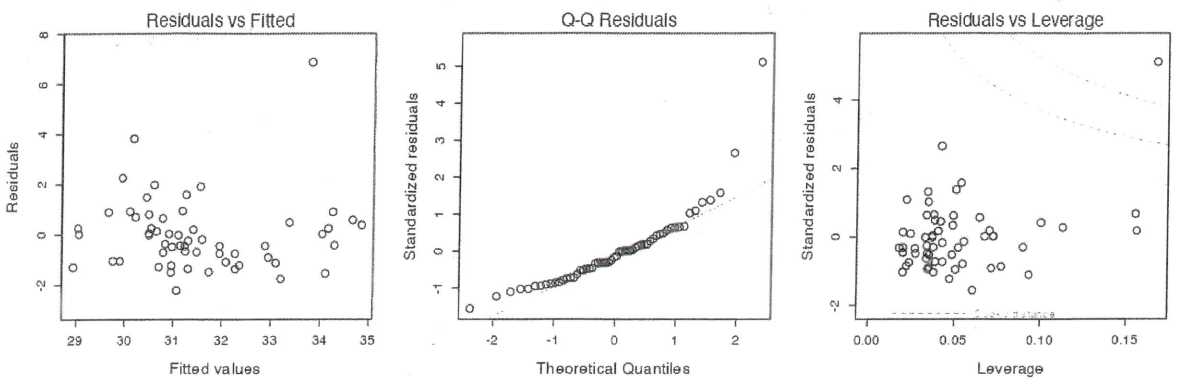
(E) There are no unduly influential points

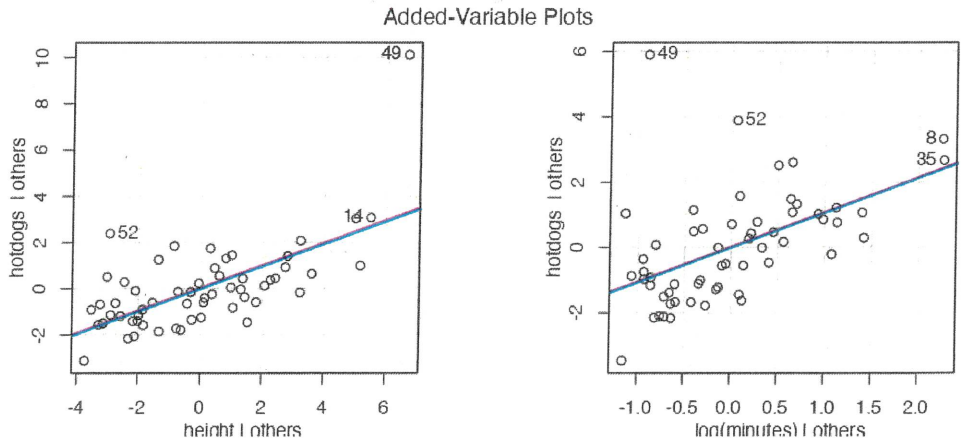
There is one point that is a bit extreme falling outside Cook's distance, but other than that it is ok

## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| log(`minutes`) | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

not necessarily, while a non significant Breusch-Pagan test can provide evidence of a better model, looking at the Plots best used for Equal variene would also be helpful in creating a mor linear model to work with.

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Model 2, while its disadvantages include a significant BP test compare to model 1, and a slight decrease in Skewness on normality, ultimately has better linearity, especially as we look at the AV plot for Hotdogs| others vs log(minutes)| others

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context

1.06702

holding all else constant for a 106% increase in minutes practed we can expect to see an incres of 1 hotdog eaten during the 10 minutes on average

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$-6.05589 + 0.48834(68) + 1.06702\log(79) = 31.81304$

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error? $n-p-1 \quad 58-2-1 = 55$

(A) 55

(B) 57

(C) 56

(D) 48

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$31.81304 \pm 2.004(1.485)$

$(28.8371, 34.79)$

# Part III - Application

## Questions 20-24 concern the following dataset:

In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

| Variable | Description |
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
foot_length   -0.21499    0.10733  -2.003   0.0507 .
ear_conch      0.15730    0.10231   1.538   0.1306
eye            0.43736    0.22549   1.940   0.0582 .
chest          0.01883    0.18154   0.104   0.9178
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
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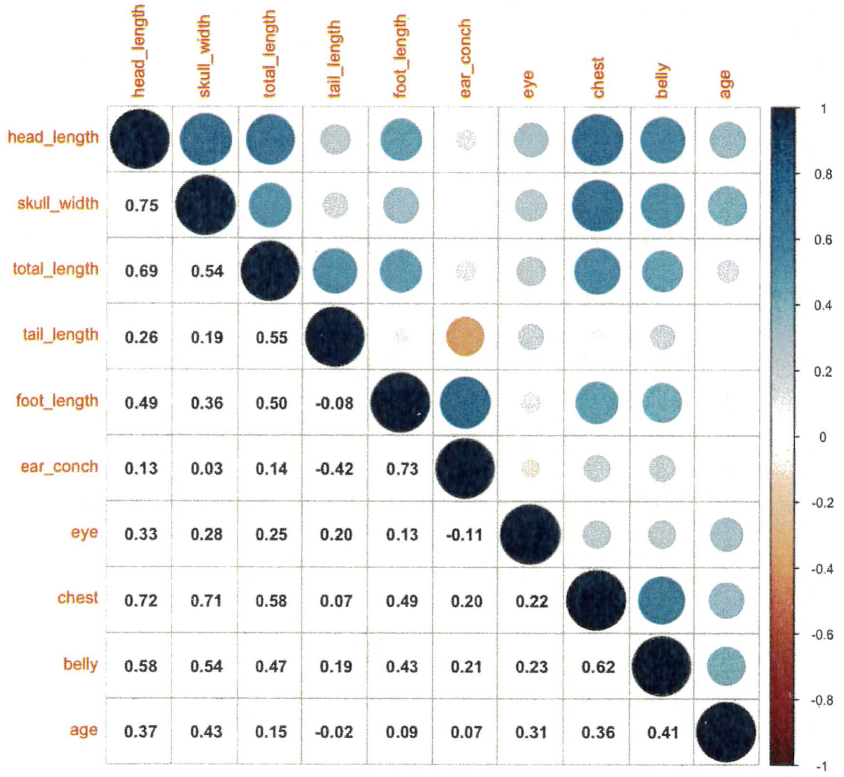
A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$0.16559 \pm 2.01(0.10599)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the true mean $\beta_{belly}$ lies between (-0.023, and 0.443

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

What the F test is showing, is that at least one of the Predictors useful in predicting our response variable, our p values are determining if our result is statistically significant compare to a hypothetical $H_0$ that we might have selected.

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

best subsets uses AIC because it is running through every possible combination, and AIC is best for Prediction, it will evaluate with a penalty of $K \cdot 2$ where $K = P+2$ (or $P+1$) to find which value is the smallest based on the model provided

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

B Average the predictions and record their value

D Choose the model with the largest RMSE

D Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

B Initialize vectors for each model that you will use to store their RMSEs

B Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

# STAT 330-001

40142034 48

## Messenger, Emma

736527038 emess 385-985-7003

Test: 640721 Form: A Wadsworth, Spencer Gordon

Entry: 04:21pm 10/21/2026 Checked out:

Employee: Jemma C.

**SPECIAL INSTRUCTIONS**

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

## PLEASE TURN OFF YOUR CELL PHONE

40142034

048

# STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

**Instructions**

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

## PART I

**Question 1: (Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

**Question 2: (TRUE/FALSE)** If the relationship between $x$ and $y$ is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

**Question 3: (Circle one answer)** When adding an additional variable to a linear regression model, the model's $R^2$ ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

**Question 4: (Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination ($R^2$) to become negative.

**Question 5: (Circle all that apply)** Which of the following are true statements about multiple linear regression (MLR)?

(A) ~~BIC will~~ usually favor a model with more predictors than the AIC criterion when performing model selection. BIC favors less predictors

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the $x$'s or $y$ may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of $\beta$ can be obtained by fitting a simple linear regression for each $x_k$ only.

(E) The assumptions on the error variables (i.e., the $\epsilon_i$) in a multiple linear regression model are exactly the same as they are for simple linear regression. $e_i = y_i - \hat{y}_i \not\Leftrightarrow E = y - X\hat{\beta}$

**Question 6: (Circle one answer)** What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model. fit w/ training data, predict w/ test data!

(C) ~~CV ensures model~~ assumptions are met which eliminates a time-consuming step of the modeling process.

(D) ~~CV is~~ the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

**Question 7: (Circle all that apply)** In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) ~~Their~~ coefficient estimates are less biased than those of OLS they introduce bias

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

Because they shrink the coefficient values using different types of penalties. Ridge uses L2 penalty, $\beta_j^2$, which shrinks the coefficients close to zero. Lasso uses L1 penalty, $|\beta_j|$, which can shrink the coefficients exactly to zero. Both Ridge and Lasso penalty's strength are controlled by $\lambda$ (lambda), as $\lambda$ increases, shrinkage becomes stronger.

Ridge: $\min\left(SSE + \lambda \sum_{j=1}^{p} \beta_j^2\right)$ — $\lambda$: shrinkage strength; $\beta_j^2$: penalty type

Lasso: $\min\left(SSE + \lambda \sum_{j=1}^{p} |\beta_j|\right)$ — $\lambda$: shrinkage strength; $|\beta_j|$: penalty type

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$D_i = \frac{(2.4)^2}{3+1}\left(\frac{0.25}{1-0.25}\right) = 1.44(0.33) = \boxed{0.48}$

$\frac{4}{p} = \frac{4}{100} = 0.04$

**(Circle one answer)** Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model. You shouldn't automatically exclude it if it exceeds the baseline.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) ~~$D_i = 0.2$~~. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) ~~$D_i = 0.2$~~. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) ~~$D_i = 0.2$~~. The observation is not influential because its leverage is less than 0.50.

(G) ~~Cook's distance cannot be calculated without knowing the observation's original residual.~~

**Question 10:**

(A) What is the formula for $\widehat{\beta}$?

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

(B) What is the conceptual difference between $\widehat{\beta}$ and $\beta$?

$\beta$ is the slope, or the relationship between a specific predictor and the response based on the true population.

$\hat{\beta}$ is our best estimate for the relationship between predictor and response based on our sample data. It is our "fitted guess" of the true $\beta$.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

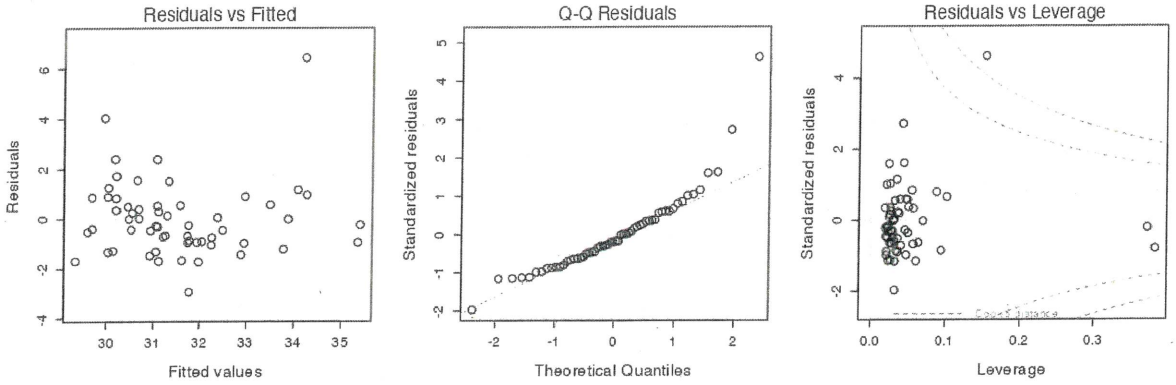
The coefficient values that produce the smallest SSE are our best estimates for $\beta$ coefficients, the smaller the SSE the more precise/accurate our model is. We can use the least squares then to find the best estimates of our $\beta$ regression coefficients.

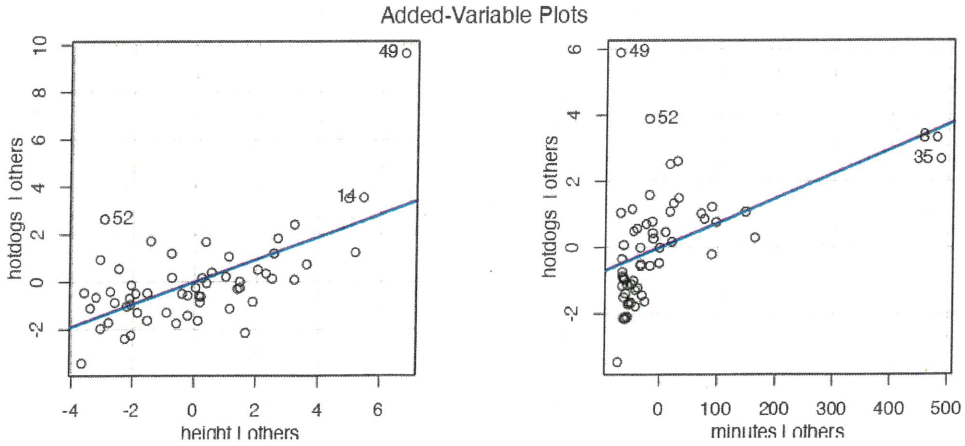
# PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

0.472892

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

$\hat{\beta}_1 = 0.472892$

A 1-unit increase in X(height) is associated with a 0.472892 unit increase in the predicted mean response of # of hotdogs eaten, holding all other predictors constant.

**Question 12:** **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0

(B) The 95% confidence interval of $\beta_2$ does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

The residual vs fitted plot shows a slight funnel shape, violating the linear assumption by not being completely randomized or patternless.

(B) There is independence between each observation

The data is fairly independent with a moderate sample size of 58 and no repeat measurments from the same individual. However, contestants were mainly selected from the 4 largest counties in the state which may create some unknown bias/influence.

(C) The errors are normally distributed

The errors are not normally distributed. The QQ plot of the residuals shows a clear curvature and departure from the normal line. While the shapiro-wilk test has a small p-value of $2.48e^{-8}$ it does not have enough power alone to prove normality.

(D) The errors have constant variance (homoscedasticity)

The Breusch-Pagan test resulted in a p-value of 0.071, which is above our alpha level of 0.05, meaning there is insufficient evidence to reject the null hypothesis, so we conclude that the errors have constant variance.

(E) There are no unduly influential points

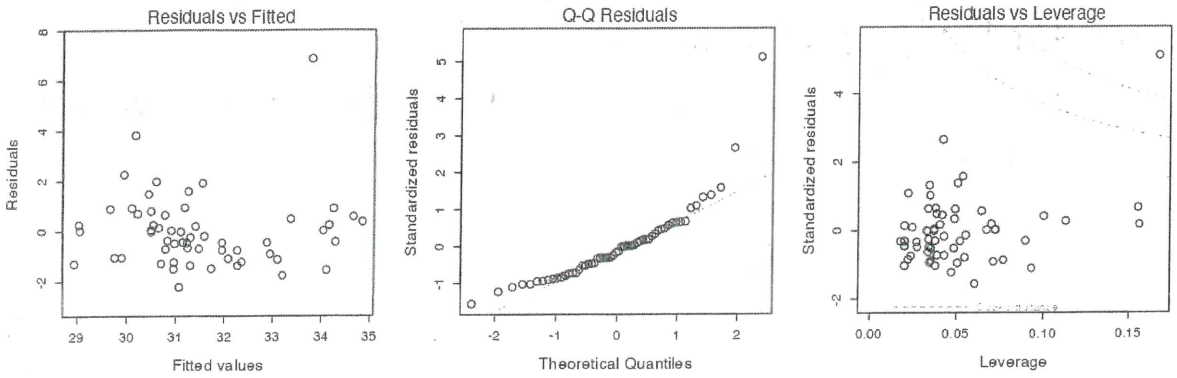
The added variable plot for height | others shows an outlier 49.
The " " for minutes | others shows an influential point 49 as their is unusual y value relative to X fit. This graph also has points, 8 and 35, with high leverage as they exhibit unusual X behaviors.

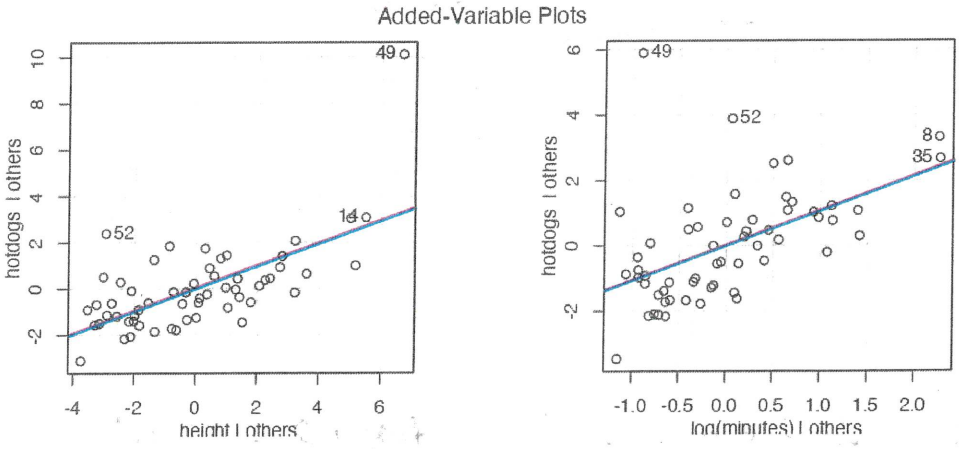
## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

level — log

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| log(`minutes`) | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

A BP test has the assumption that our null hypothesis is that our errors have constant variance. A "superior model" would have a constant variance in our errors. So, we don't want to reject Ho, we want a high p-value. Model 1 is the superior model since it's p-value is high and we fail to reject Ho. Model 2 has a low p-value, leading us to reject Ho, our unwanted result.

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

There is still a slight pattern in our residual plot (linearity), independence still holds, the QQ plot is still not lined up (normality violated) and the Breusch Pagan test favors model 1 over model 2. The added variable plots show less influential points in model 2 rather than model 1.

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context 1.067

$\hat{\beta}_2 = 1.067$ $+1\%\ X \Rightarrow \sim \beta\%$ in median $y$

For every 1% increase in X (minutes), about at 1.067% increase in the median y (hot dogs eaten) response is expected, holding all other predictors constant.

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$y_i = \beta_0 + \beta_1 \text{height}_i + \beta_2 \log(\text{min.}) + \epsilon_i$

$y_i = -6.05589 + 0.48834(68\text{in}) + 1.06702(4.369) = 31.8 \text{ hotdogs}$

33.20712 4.6618

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error?

$n-p-1$

$58-2-1 = 58-3 = 55$

(A) 55

(B) 57

(C) 56

(D) 48

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

prediction!

PI: $\hat{y}_0 \pm t^* s\sqrt{1+\cdots}$

$31 + 2.004(1.485) = 34.77$

$31 - 2.004(1.485) = 28.02$

2.97594

PI = 28.02 - 34.77 hot dogs

# Part III - Application

## Questions 20-24 concern the following dataset:

In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

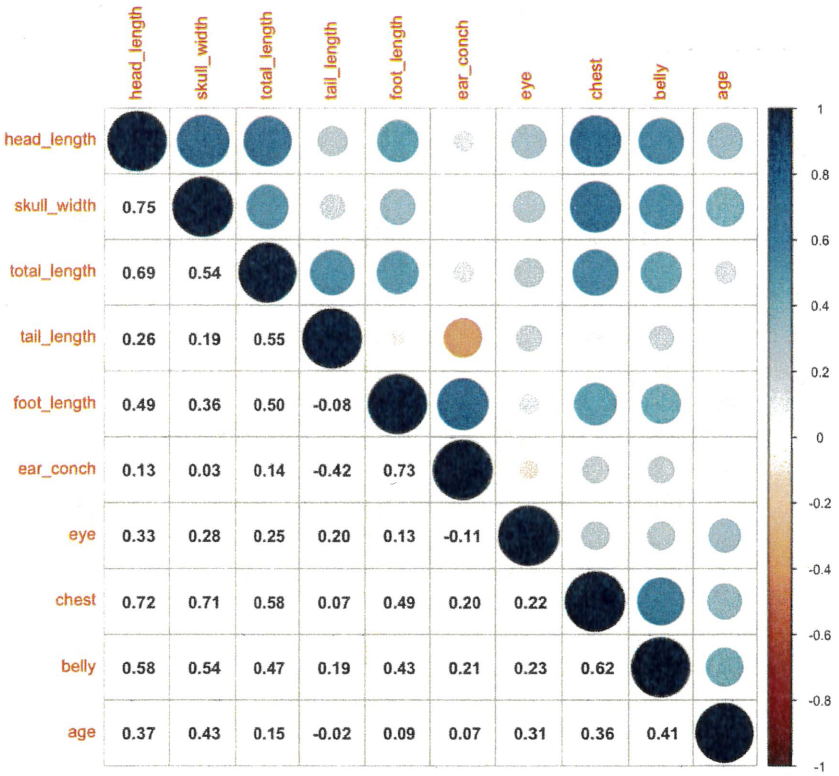
| Variable | Description |
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

10 predictors

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.




$VIF = \frac{1}{1-R^2}$

**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

0.16559

$CI = \hat{\beta}_i \pm t^* \times SE(\hat{\beta}_i)$

$0.16559 \pm 2.01 \times 0.10599$

$0.16559 \pm 2.01 \times 0.10599$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the true population mean for belly width among local Australian possums lies in the interval (-0.023, 0.443) cm, holding all other predictors constant.

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

There is some kind of interaction between 2 or more of the predictors that, on their own, is not present but when they interact affects our F-test results.

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

AIC penalizes extra predictors lightly as it determines that better fit models out weight the penalty of additional predictors. Best subsets algorithm can search for every possible combination of predictors, not restricted by a path-based model. It compares each possible predictor combo to it's AIC value, looking for the lowest AIC and therefore the best fit model. However, AIC allows for models with lots of predictors and small AIC values, and the best subset can test any combo of predictors, no matter how big, so it may result in selecting a model of best fit with lots of predictors still included.

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

adj. $R^2$ penalizes complex models so models with 6, or 7, or more predictors than this selected model, 5 predictors, will have higher adj. $R^2$ values.

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop → set up

(B) within the for loop → in the training

(C) after the for loop → interpret/conclude

(D) never in the CV process

Steps:
1) split
2) Fit to training data
3) Predict test data
4) record test error
5) repeat
6) summarize
* never fit on test data! *

C Average RMSE across all test-training splits and report

B Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

B Compute each model's AIC and BIC

D Fit each model using the testing data

A Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

C Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM